

Internship report

Ramp-Guided Hydrogel Actuation for Repeatable Microfluidic Compression

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Abstract

This internship developed a microfluidic method for laterally compressing confined hydrogel cylinders while maintaining optical access. Pneumatic deformation of a PDMS ceiling was converted into lateral motion through a ramp-guided hydrogel pusher. Several geometries were tested before selecting a one-sided design combining a single pusher, a fixed wall, and local vertical confinement. This configuration allowed successive cylinders to be compressed using the same chip and pusher. Their relative mechanical responses were compared using direct image-edge measurements of horizontal strain, with pusher deformation serving as an uncalibrated loading coordinate. The device enabled repeatable compression and a conditional comparison of mechanical response, although tracking uncertainty, experimental variability, the absence of direct force measurements, and the lack of replicated specimens at each exposure dose prevented the determination of absolute material properties or a statistically resolved stiffness ordering.

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1 Introduction

1.1 Scientific context

Mechanical cues influence the organisation, function, and evolution of biological materials [10]. Consequently, there is a continuing need for experimental tools that can impose controlled loads on soft three-dimensional samples while preserving optical access. Microfluidic devices are well suited to this task because they confine small volumes, allow fluid exchange, and can be integrated directly with microscopy. Polydimethylsiloxane (PDMS) is particularly useful: it is transparent, biocompatible for many laboratory applications, easy to mould, and can serve as both the body and the actuator of a device [5, 7, 6].

Jain *et al.* introduced a monolithic PDMS device in which a sample chamber is separated fluidically from adjacent pneumatic cavities. Applying negative pressure to the air cavities moves the chamber ceiling downwards [1]. Later work at LadHyX showed that the magnitude and spatial profile of this ceiling motion are determined by the global chip geometry: the same fabrication principle can produce several deformation modes [2]. Mathis Ratte subsequently used photopatterned hydrogel structures to characterise the ceiling motion and the deformation of confined gels [3].

For this internship that vertical actuation principle is extended by adding a ramp to the central channel. A movable hydrogel block is photopolymerised beneath the inclined ceiling. When the ceiling is lowered, the ramp converts part of the vertical motion into a lateral motion of the block, directed towards the region of greater channel height. The block can therefore act as an in-chip pusher. The target object in this study was a photopolymerised hydrogel cylinder located in the same channel.

1.2 Problem statement

The first difficulty was not simply to produce motion, but to obtain the same useful contact geometry repeatedly. A free cylinder can translate, roll, or escape transversely when approached by one or two compliant blocks. A successful device therefore had to satisfy four requirements:

- convert pressure-driven ceiling displacement into a sufficiently large lateral stroke;
- guide the pusher towards the cylinder without permanent bonding between the two;
- prevent the cylinder from escaping during loading while keeping it optically visible;
- permit flushing and replacement of the cylinder so that several samples could be tested with the same chip and the same pusher.

The hydrogel cylinders served as controlled surrogates for cellular spheroids. Their simple geometry made it possible to develop the actuation and analysis methods before considering heterogeneous biological samples. Once repeatable contact had been obtained, the second objective was to compare cylinders of different stiffness. The intended output was a relative ranking within one chip rather than an absolute Young's modulus, because neither the applied force nor the evolving three-dimensional contact state was measured. Reusing one pusher and one chip suppresses two major sources of variation: the block geometry and the chip-specific relation between pneumatic pressure and ceiling displacement. Cylinder geometry, loading rate, contact state, and image-analysis quality remain important and are treated explicitly below.

2 Methods and materials

2.1 Microfluidic devices

2.1.1 Initial two-ramp chip

The first experiments used an existing mould designed by Alexandre Avaro. The resulting device contained four pneumatic air chambers and one elongated central microfluidic channel. The channel ceiling comprised two opposing ramps separated by a flat central region. Fluid could be injected through two access holes located at the two ends of the central channel, which also allowed unpolymerised precursor to be flushed out.

All glass supports used during this internship were coated with a thin PDMS layer before chip assembly. This changed the surface presented to the hydrogel from glass to PDMS. The purpose was to suppress the strong adhesion observed when PEG-based structures are polymerised directly on glass and thereby allow the pusher blocks and cylinders to move.

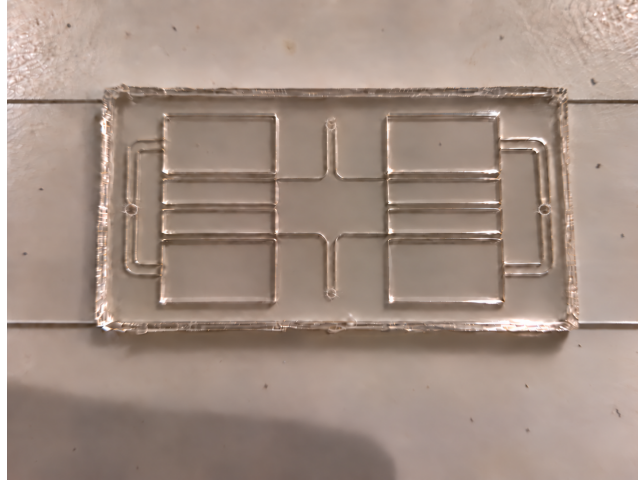
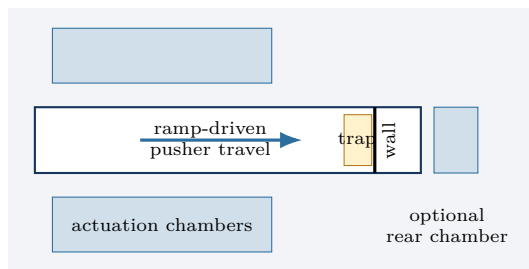


Figure 1: Initial chip. Top-view photograph showing the four air chambers, the elongated channel, the two ramps and the central flat section

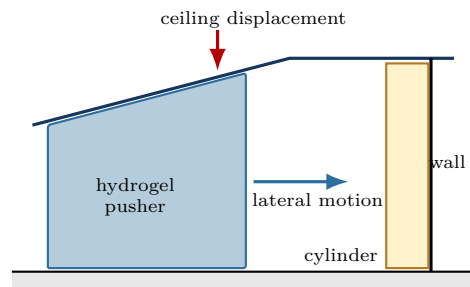
2.1.2 Redesigned one-sided chip

The final design was produced in Autodesk Fusion. Conceptually, it retained one ramp and the adjacent flat section of the initial chip, but terminated the compression region with a PDMS wall. The cylinder was pushed against this wall by a single ramp-driven block. A local height feature was added next to the wall to confine the cylinder through the 0.6 mm channel thickness and favour its placement in the loading zone. It did not impose complete in-plane confinement, so lateral drift remained possible. An additional pneumatic chamber was placed behind the wall with the initial intention of lowering the ceiling above the cylinder; its role and eventual rejection are discussed in [section 4.4](#).

The CAD geometry was transferred to an aluminium master by CNC milling. This fabrication route was selected to reproduce the ramps and slanted surfaces more accurately than the available alternatives.



(a) Conceptual top view.



(b) Conceptual longitudinal section.

Figure 2: Operating principle of the one-sided device. The drawings are schematic and not to scale.

2.1.3 PDMS casting, surface preparation, and bonding

The standard fabrication protocol followed the method used for the preceding LadHyX devices [3, 2]. Base and curing agent were mixed at a 10:1 mass ratio, poured into the master, degassed under vacuum for approximately 30 min, and cured for 2 h at 70 °C. After curing, the PDMS device was peeled from the master and access holes were punched for the fluid ports and the pneumatic connections. The device and PDMS-coated glass support were cleaned with ethanol and adhesive tape, plasma-activated for approximately 45 s, brought into contact, and heated at 100 °C for 5 min to reinforce bonding. The coating on the glass was prepared by spin-coating PDMS at 1000 rpm for 30 s, followed by curing at 70 °C. Formulation and curing history were kept fixed because both affect the mechanical response of PDMS [4].

2.2 Hydrogel formulation and photopatterning

2.2.1 Formulation screening

Hydrogel structures were made from polyethylene glycol (PEG), polyethylene glycol diacrylate (PEGDA), and lithium phenyl-2,4,6-trimethylbenzoylphosphine (LAP). PEGDA forms the covalently crosslinked network under illumination, PEG modifies the liquid formulation and the resulting network, and LAP initiates the photopolymerisation. PEG-based hydrogels are attractive because their formulation and exposure can be varied over a broad mechanical range [9, 8].

Several PEG/PEGDA/LAP proportions were tested. Some compositions did not polymerise reliably, whereas other trials produced excessive or spatially extended polymerisation. A second operational failure occurred occasionally when many shapes were patterned in a single PRIMO operation: unintended over-polymerised regions appeared in the channel. This was treated as a printing failure rather than as usable material data. The formulation ultimately used for the device-development and comparison experiments was 60 % PEG, 25 % PEGDA, and 15 % LAP, expressed as fractions of the laboratory stock solutions.

2.2.2 In-chip photopolymerisation

The precursor was injected through one fluid port until the central channel was filled and air bubbles had been removed. Shapes were projected with the Alveole Lab PRIMO system. The pusher block was printed on the ramp at 0.7 mJ mm^{-2} , and each cylinder was printed next to the terminal wall. The PRIMO exposure dose controls photochemical conversion and crosslink density and can therefore modify the apparent cylinder stiffness. The plain cylinder series used doses of 0.6 mJ mm^{-2} , 0.7 mJ mm^{-2} , 0.8 mJ mm^{-2} , and 0.9 mJ mm^{-2} . A second 0.6 mJ mm^{-2} cylinder contained internal horizontal stripes and was analysed separately as a structurally reinforced sample.

To reduce unintended polymerisation, the masks were assembled from the basic shapes supplied by the PRIMO software without rescaling or rotation. Modified primitives were sometimes projected in several steps, causing repeated illumination of overlapping regions and local over-polymerisation.

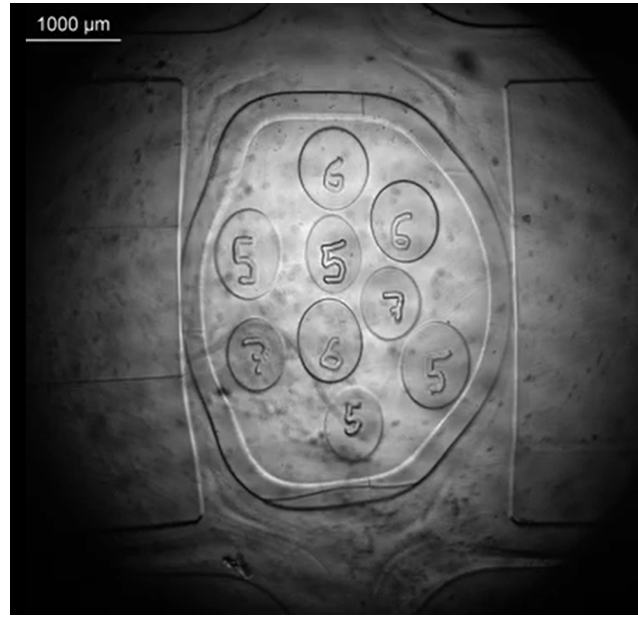


Figure 3: Example of over-polymerisation where we tried polymerising with different exposures small circles inside a hollow circle : the hollow circle is being compressed without the circles inside it getting in contact.

2.3 Pneumatic actuation, flow, and imaging

The microfluidic and pneumatic circuits were independent. Fluid was injected into the central channel through the two end ports using a pipette or a syringe when continuous flow was needed. The air chambers were connected to a pressure controller. For the comparison series, the programmed pressure was decreased linearly from 0 mbar to -900 mbar and subsequently released. The command was nominally the same for every run, although the realised chip deformation and contact history were not independently measured and were not perfectly reproducible.

Compression was observed in bright field using a Nikon microscope controlled with NIS-Elements, following the imaging workflow of the preceding study [3]. Videos were acquired with a $2\times$ objective at either 5 or 10 frames per second.

2.4 Same-chip cylinder-comparison protocol

To reduce device-to-device variability, all cylinders in one comparison series were tested in the same chip with the same pusher block. The procedure was:

1. Fill the bonded chip with the selected 60/25/15 precursor and print one pusher block on the ramp using a fixed mask and a dose of 0.7 mJ mm^{-2} .
2. Print a cylinder of prescribed diameter and dose in the height trap next to the wall.
3. Apply the decompression while recording the full loading and unloading sequence. Keep illumination, focus, and frame rate fixed.
4. Release the pressure completely. Inject liquid through the channel and push on the block until the tested cylinder is flushed out through the bottom fluid port, while retaining the original pusher.
5. Print the next cylinder and repeat the same acquisition.

An equal-dose reference cylinder was not repeated at the end of the series. In retrospect, a reference cylinder before and after the sequence would have provided a direct measurement of pusher or chip drift. This within-chip design holds the pusher geometry, local trap, channel height, wall, and pneumatic coupling nearly constant. It does not eliminate changes caused by repeated UV exposure, ageing or swelling of the pusher, or small differences in cylinder diameter. These effects are included in the uncertainty discussion in [section 5.6](#).

3 Theory

3.1 Why the loading condition matters

Stiffness is a relation between force and deformation; a displacement measurement alone is not a material property. In the present device, the pressure controller prescribes the pneumatic pressure difference Δp . This deforms the monolithic PDMS ceiling, and the ramp converts part of that motion into a lateral pusher stroke. The controller therefore sets an *actuator state*: it sets neither the force on the cylinder nor the cylinder displacement directly. Once contact is established, the available motion is shared between deformation of the pusher, the cylinder, and the surrounding chip.

This distinction can be illustrated with a minimal one-dimensional model. Let $s_0(\Delta p)$ be the pusher's free lateral stroke at a given pressure, k_a the incremental stiffness of the pusher–chip actuator, K_i the effective lateral stiffness of cylinder i , and δ_i its horizontal compression. Linearising locally about one loading state gives

$$F_i = K_i \delta_i = k_a (s_0 - \delta_i), \quad \delta_i = s_0 \frac{k_a}{k_a + K_i}, \quad F_i = s_0 \frac{k_a K_i}{k_a + K_i}. \quad (1)$$

At the same pressure command, the transmitted force F_i generally depends on the cylinder stiffness; equal pressure is therefore not equivalent to equal force. Nevertheless,

$$\frac{\partial \delta_i}{\partial K_i} = -s_0 \frac{k_a}{(k_a + K_i)^2} < 0, \quad (2)$$

so a more compliant cylinder deforms more if s_0 and k_a are reproducible. This model is used only to justify an ordinal comparison. It does not describe the large-deformation, frictional, three-dimensional contact in sufficient detail to identify a material modulus.

3.2 Cylinder strain and relative stiffness

Let $D_{0,i}$ be the unloaded diameter of cylinder i , measured from its own image sequence, and let $D_{x,i}(t)$ be the direct horizontal span of its visible boundary. After wall contact, the wall-facing endpoint is bounded by the detected rigid wall and the opposite endpoint is taken from the pusher-facing cylinder edge. The positive horizontal compression strain is

$$\varepsilon_{x,i}(t) = \frac{D_{0,i} - D_{x,i}(t)}{D_{0,i}}. \quad (3)$$

Normalising by the diameter removes the leading effect of the small size differences between cylinders. It does not remove all geometric effects, because curvature, confinement, and contact position can still change with diameter. The use of the direct edge span is essential: an ellipse fit would impose a shape that the flattened cylinder need not obey.

At a genuinely common force F^* , a secant effective stiffness would satisfy

$$K_i^{\text{eff}} = \frac{F^*}{D_{0,i} \varepsilon_{x,i}}, \quad \frac{K_i^{\text{eff}}}{K_j^{\text{eff}}} = \frac{D_{0,j} \varepsilon_{x,j}}{D_{0,i} \varepsilon_{x,i}}. \quad (4)$$

For nearly equal initial diameters, smaller strain at the same force therefore means a stiffer effective response. Equation (4) is not used here to report a modulus ratio, because the force was not measured and the evolving three-dimensional contact area, confinement, and friction are unknown. The elementary uniaxial relation $E = F/(A\varepsilon)$ is consequently inapplicable.

The pressure history alone is not an adequate comparison coordinate, as shown by [equation \(1\)](#). Instead, the analysis uses the deformation of the same physical pusher as an uncalibrated loading coordinate. Let $x_b(y, t)$ be the detected pusher surface and $x_{\text{ref}}(y, t)$ the far-field surface that would be present without the local cylinder indentation. With the sign chosen so that indentation is positive,

$$u_b(y, t) = [x_b(y, t) - x_{\text{ref}}(y, t)]_+, \quad (5)$$

and the dimensionless normalised indentation-area coordinate is

$$\eta_b(t) = \frac{1}{D_{0,i}^2} \int_{\Gamma_b} u_b(y, t) \, dy. \quad (6)$$

The symbol η_b denotes a geometric deformation measure, not a force or a force per unit length. Because the same physical pusher is used throughout the series, η_b is assumed to vary monotonically with transmitted load on the loading branch. Matching η_b is therefore more informative than matching frame number or commanded

pressure alone. It is still not force matching: changes in contact position, pusher history, or deformation shape can alter the relation between η_b and F .

Each strain curve is interpolated over the pusher-deformation interval shared by all videos, $\eta_b \in [\eta_1, \eta_2]$. The mean strain over this matched-pusher interval and the reported relative index are

$$\bar{\varepsilon}_{x,i} = \frac{1}{\eta_2 - \eta_1} \int_{\eta_1}^{\eta_2} \varepsilon_{x,i}(\eta_b) d\eta_b, \quad I_{\varepsilon,i} = \frac{\max_j \bar{\varepsilon}_{x,j}}{\bar{\varepsilon}_{x,i}}. \quad (7)$$

The most deformable sample has $I_{\varepsilon} = 1$, and a larger value corresponds to a smaller mean horizontal strain within this specific experiment. The complete $\varepsilon_x(\eta_b)$ curves remain the primary result: I_{ε} depends on the chosen averaging interval and is neither a modulus ratio nor, by itself, evidence of a statistically resolved stiffness difference. If two curves cross, their ordering is load-dependent and a single unconditional rank is not supported.

4 Experimenting with different designs

4.1 Two opposing blocks in the initial chip

The first strategy used the symmetry of the two-ramp chip. One hydrogel block was printed on each ramp and a free cylinder was printed in the middle of the channel. Negative pressure in the four air chambers moved the two blocks towards the centre. Although both blocks moved, the cylinder repeatedly escaped instead of remaining between them. Once the cylinder acquired a transverse displacement the blocks tended to expel it.

This trial established an important design principle: two-sided actuation does not by itself produce self-centering. Repeatability requires a separate geometrical constraint on the target, not only symmetric forces.

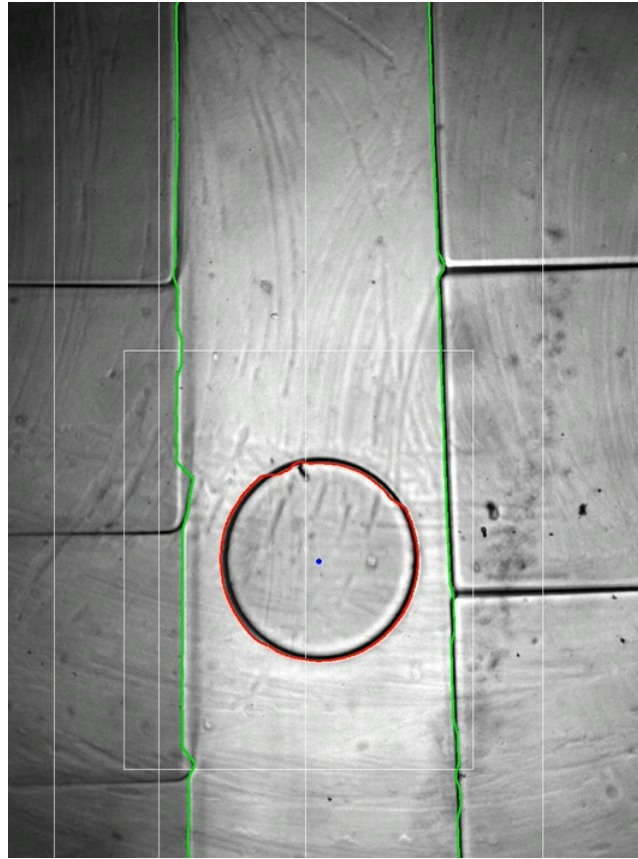
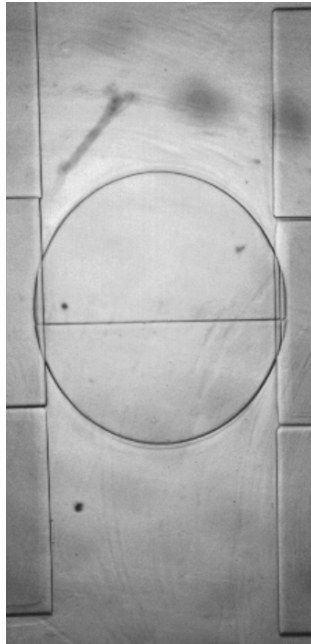


Figure 4: Initial two-pusher experiment.

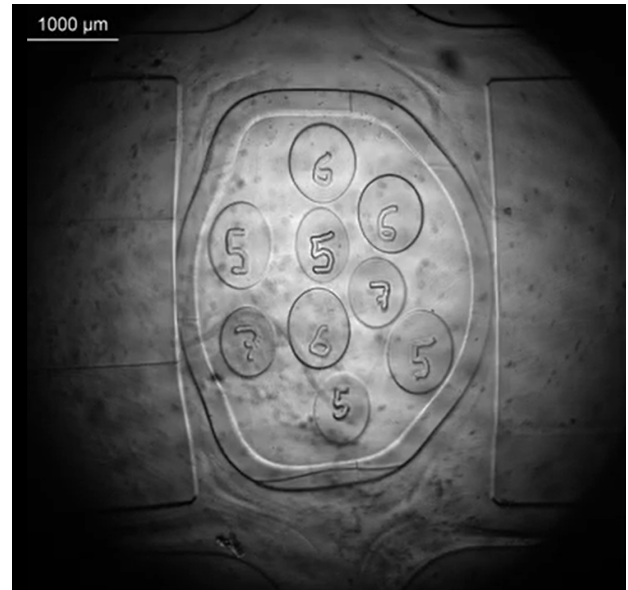
4.2 Bonded composite and ring demonstrations

In one trial the central cylinder was photopolymerised while connected to both blocks. A clear compression video was obtained. This confirmed that the pressure-driven ramps could generate relative motion and deform a central gel, but it was not a valid test of free contact: the polymerised bridges transmitted shear and tension in addition to compression, fixed the cylinder position, and altered the effective stiffness of the assembly. The video is therefore useful as a qualitative actuator demonstration and was excluded from relative-stiffness analysis.

Another exploratory print connected a large circular ring between the two blocks and placed many small cylinders inside it. Liquid trapped inside the ring polymerised accidentally. This one-off trial was not designed as a controlled experiment and yielded no quantitative result. It nevertheless illustrated why enclosed masks and large simultaneous projections are susceptible to unintended polymerisation and why the final protocol limited the number of shapes printed at once.



(a) Cylinder printed bonded with both blocks



(b) Large ring with internal cylinders and accidental polymerisation.

Figure 5: Qualitative bonded demonstrations. These images document design exploration and are not used for stiffness measurement.

4.3 Interlocking pusher geometries

The next approach modified the block outlines so that one block fitted partly into the other, with the aim of surrounding and retaining the cylinder during approach. Several geometries were tested. Occasional compression could be seen, but the cylinder still slipped away and the videos were not sufficiently repeatable for quantitative use.

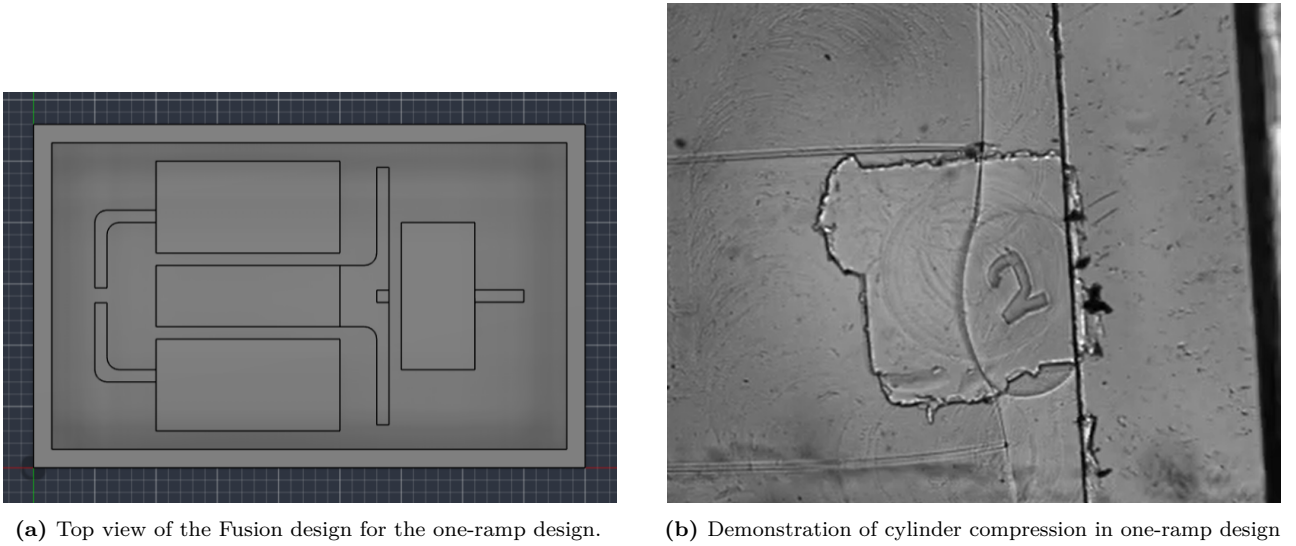
4.4 One-sided compression against a wall

The repeated escape in the symmetric design motivated a change from two moving boundaries to one moving boundary and one fixed reaction wall. The new Fusion design can be understood as one half of the original chip: one ramp generates pusher motion, a flat section provides the working region, and a PDMS wall terminates the channel. This architecture removes the need to synchronise two blocks and fixes the direction of loading.

The wall alone was not sufficient, because a cylinder could still leave the intended contact zone. A local ceiling-height feature was therefore added adjacent to the wall. This feature maintained vertical confinement and biased the cylinder towards the loading region without fully fixing its in-plane position. The open fluid path still allowed the cylinder to be removed after pressure release. In the tested device, the combination gave sufficiently repeatable positioning and compression for the comparison experiments, although some vertical motion in the image plane remained visible.

An extra air chamber was included behind the terminal wall. The initial idea was that its depressurisation would lower the ceiling locally above the cylinder and help hold the sample in place. No useful effect was observed, and the chamber was not used in the final experiments.

This negative result is mechanically consistent with the final geometry. First, the ceiling displacement is generally reduced close to a channel boundary where the surrounding PDMS is constrained. Second, the cylinder sits directly against the wall, whereas the greatest displacement induced by a nearby chamber can occur farther from that boundary. Third, a vertical force would increase normal contact but would not by itself create an in-plane restoring force. The passive local confinement feature was therefore simpler and more useful.

**Figure 6:** One sided compression chip.

4.5 Design selection

Table 1: Summary of the design iterations.

Design	Independent cylinder	Prevented escape	Flushable	Decision
Two opposing rectangular blocks	Yes	No	Yes	Rejected: unstable positioning
Cylinder bonded to both blocks	No	Yes	No	Actuation demonstration only
Interlocking pusher outlines	Yes	Inconsistent	No	Rejected: cylinder still escaped
One pusher, wall, and local confinement	Yes	Yes	Yes	Selected

The selected design solved the primary engineering objective: it converted ceiling motion into a directed pusher stroke and maintained a free cylinder near a defined wall contact often enough to perform a comparison series. The next section evaluates both the information obtained from that series and the residual lack of repeatability.

5 Experimental validation of relative stiffness

5.1 Validation strategy

The selected device was validated at two levels. The first was mechanical repeatability: the pusher had to reach a confined cylinder, compress it against the same wall, and release it without escape. The second was discriminatory ability: cylinders made with different photopolymerisation conditions had to produce distinguishable deformation under a common actuation protocol.

The hydrogel formulation was held at 60/25/15 for all cylinders in the comparison series. Plain cylinders were exposed at 0.6 mJ mm^{-2} , 0.7 mJ mm^{-2} , 0.8 mJ mm^{-2} , and 0.9 mJ mm^{-2} ; the pusher was exposed at 0.7 mJ mm^{-2} . A second 0.6 mJ mm^{-2} cylinder contained internal horizontal stripes. Cylinder mask, channel location, liquid, and commanded pressure history were kept as similar as possible. The cylinders were nominally equal in diameter within about $\pm 20 \mu\text{m}$; the edge-derived initial diameters nevertheless ranged from approximately 242 to 259 pixels, reflecting both fabrication and segmentation differences. After each video, the old cylinder was flushed away and a new one was printed, while the chip and pusher block were retained. This protocol reduces, but does not eliminate, variations in pusher shape, local confinement, contact position, and device compliance.

5.2 Video-analysis workflow

The raw recordings were cropped in space and time in Fiji to retain the pusher, cylinder, rigid wall, and useful loading interval. Static background was suppressed by subtracting a temporal median, and an edge-enhanced TIFF stack was exported. These operations improve boundary visibility but also remove intensity information

Table 2: Descriptive direct-strain results over the common normalised pusher-deformation interval $\eta_b \in [0.0138, 0.2383]$. The index I_ε is defined by equation (7); the most deformable measured sample has value 1. It is not a modulus ratio and does not establish a complete statistical ranking. The striped cylinder is shown for completeness but excluded from the plain-cylinder dose comparison.

Sample	Dose (mJ mm ⁻²)	Valid frames	$\bar{\varepsilon}_x$	I_ε	Interpretation
Plain cylinder (07_06)	0.6	16	0.2726	1.000	Consistently most deformable plain sample
Striped cylinder (07_06)	0.6	19	0.1996	1.366	Separate exploratory structural case
Plain cylinder (07_07)	0.7	26	0.2291	1.190	Not resolved from 0.8 and 0.9
Plain cylinder (07_08)	0.8	24	0.2418	1.127	Not resolved from 0.7 and 0.9
Plain cylinder (07_09)	0.9	42	0.2138	1.275	Lowest interval-mean strain among plain samples; curves cross

and can strengthen unwanted edges from the trap, wall texture, or internal stripes.

The analysis was then performed in a standalone Jupyter notebook. A circular Hough transform initialised the undeformed cylinder and a constrained ellipse tracker supplied a stable search window in later frames. Crucially, the ellipse axes were not used to compute strain. Within the localisation window, the algorithm selected a globally continuous, closed path through image edge pixels. The direct horizontal extent of this path gave $D_x(t)$; after contact, the wall-facing endpoint was bounded by the detected wall so that a fitted contour could not extend through the rigid boundary. The initial diameter D_0 was estimated from pre-contact frames when they were available. For the 0.6 plain and 0.9 recordings, which began at or very near two-sided contact, D_0 relied partly on the approximate in-plane area-conservation assumption, which introduces additional uncertainty.

A second continuous path followed the pusher surface. Portions above and below the cylinder defined a far-field reference, from which the local indentation and η_b in equation (6) were calculated. The pusher path was constrained to remain on the pusher side of the cylinder and not switch to the fixed wall. Only the loading branch after two-sided contact was retained. Frames were rejected when the direct edge support was weak, the apparent cylinder area departed excessively from its initial value, or the strain lay outside the selected moderate-deformation interval. Every retained result was checked using diagnostic overlays; automatic confidence values were treated as filters, not as proof of correct tracking.

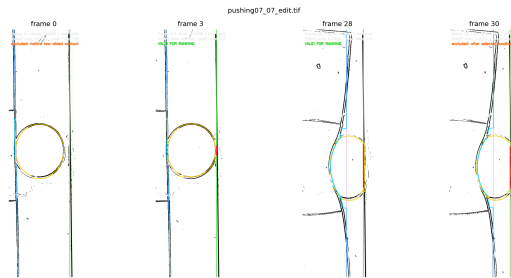


Figure 7: Tracking overlay on compression of a cylinder exposed at a dose of 0.7mJ mm⁻²

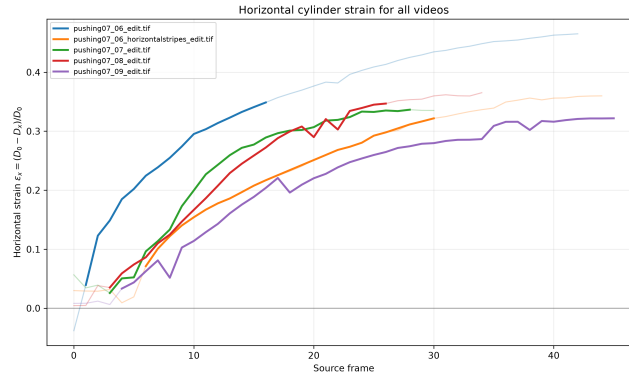


Figure 8: Horizontal strain for all videos

5.3 Repeatable cylinder compression

The one-sided device qualitatively improved repeatability compared with all earlier designs. In the selected videos, the pusher travelled from the ramp onto the flat section, made contact with the cylinder, and compressed it against the wall without complete escape. On pressure release, the geometry reopened sufficiently for the cylinder to be removed by flow and replaced while the pusher remained in the same chip. Nevertheless, the cylinder underwent visible motion in the image plane, and the pusher approached it at slightly different positions from one run to another. The experiment was therefore repeatable at the level required to obtain compression curves.

5.4 Preliminary assessment of experimental repeatability

A preliminary repeatability test was performed using three cylinders polymerized at the same nominal dose of 0.7 mJ mm^{-2} and compressed in three independently fabricated microfluidic chips. This comparison was intended to evaluate the repeatability of the complete experimental protocol, including cylinder fabrication, chip geometry, loading transmission and image analysis. Since the applied force was not measured, it does not provide a direct estimate of the repeatability of the cylinder Young's modulus or mechanical stiffness.

The measured initial cylinder diameters were 267.3 ± 6.1 pixels, where the uncertainty denotes the sample standard deviation of the three measurements. The corresponding coefficient of variation was approximately 2.3%. The horizontal compression was normalized by the initial diameter of each cylinder, thereby reducing the influence of these small geometrical differences.

Figure 9 presents the horizontal-strain histories after alignment at the first frame in which simultaneous contact with the pusher and the rigid wall was detected. The experiments differed considerably during the initial contact transient. At approximately 0.20 s after contact, the measured strains ranged from 0.024 to 0.146. This early variability may result from uncertainty in the exact contact frame, differences in the initial pusher position, variations in the fabricated chip geometry, and the small absolute deformation being measured during the first frames.

After approximately one second, the three strain curves became considerably closer. At $t = 1.5 \text{ s}$, the mean horizontal strain was 0.229, with a sample standard deviation of 0.009. At $t = 3.0 \text{ s}$, it was 0.291 ± 0.006 . These values are descriptive statistics for only three experiments, but they show that the measured strain evolution became similar during the intermediate part of the nominal pressure-control sequence.

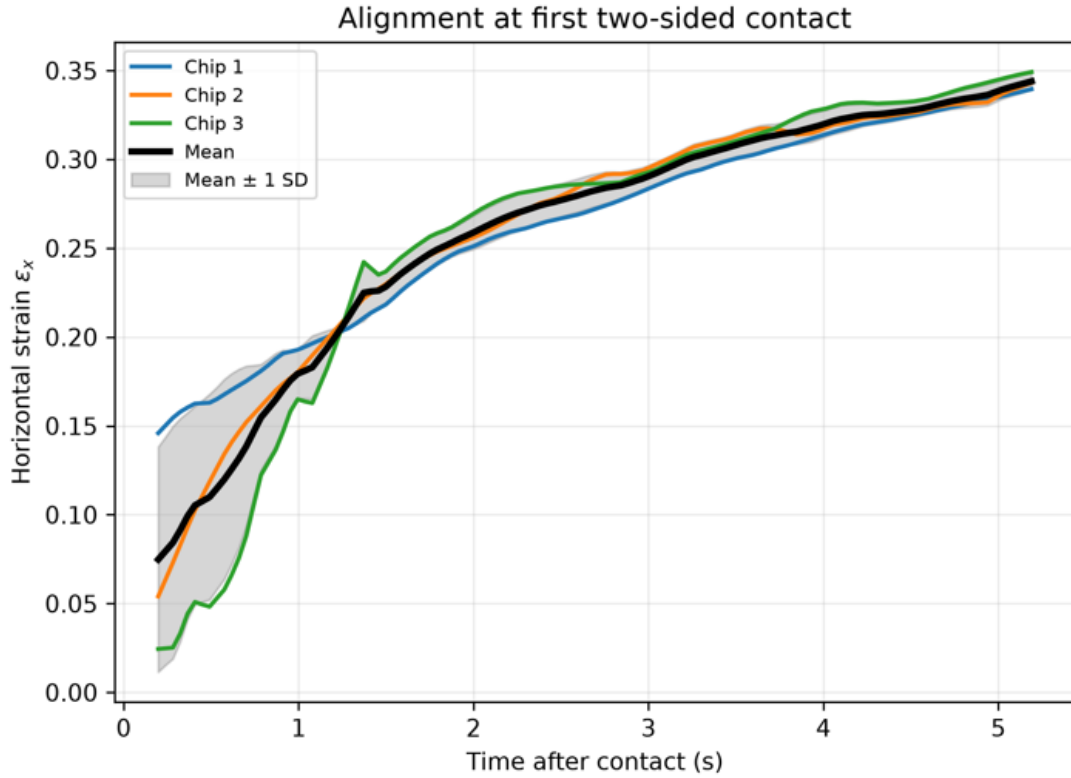


Figure 9: Horizontal-strain histories for three cylinders polymerized at 0.7 mJ mm^{-2} and tested in independently fabricated chips. Time is measured from the first detected simultaneous contact with the pusher and the rigid wall. Individual experiments are shown in colour, the black curve represents their mean, and the grey region represents the mean plus or minus one sample standard deviation. The statistics are descriptive because only three experiments were available.

These results indicate that the image-analysis method can reproduce the intermediate strain evolution of nominally identical cylinders under similar operating conditions. They also show that variability in the initial contact and loading transmission remains important. A larger number of specimens, reliable measurement of the pusher displacement, and preferably a calibrated force measurement would be required to quantify the uncertainty of the cylinder stiffness itself.

5.5 Deformation of the striped cylinder

The striped cylinder exhibited a qualitatively different deformation from the homogeneous cylinders. The stripes correspond to more strongly polymerized, and therefore stiffer, regions extending through the complete vertical thickness of the cylinder. They consequently behave as reinforcing inclusions embedded in a softer hydrogel matrix. Their curvature during compression is therefore mechanically possible: under asymmetric loading, even a single stiff stripe could bend into a U-shape as the surrounding softer material deforms.

More notably, several stripes developed an S-shaped profile during compression, as shown in Fig. 10. This observation is consistent with spatially varying in-plane shear and local rotation. Possible causes include imperfect alignment between the pusher and the cylinder, asymmetric contact with the wall, friction against the vertically confining surfaces, and local variations in the geometry or stiffness of the patterned material.

Dark regions also appeared beside some of the stripes as their curvature increased. Since the reinforced regions extend through the complete height of the cylinder, these optical features may indicate that the stripes tilted or buckled through the thickness, producing local variations in orientation, height, or optical path length. This would be compatible with a three dimensional deformation involving out of plane shear or twisting. However, the top-view images cannot distinguish this interpretation conclusively from optical effects such as refraction, defocusing, or artefacts introduced by edge enhancement. Depth-resolved imaging, such as confocal microscopy or a z-stack through the cylinder thickness, would be required to demonstrate three dimensional twisting directly.

The striped cylinder must therefore be interpreted as an anisotropic reinforced structure rather than as a homogeneous material characterized by a single effective modulus. Its lower horizontal strain indicates greater effective structural rigidity in the loading direction, while the S-shaped stripes reveal an additional shear-dominated deformation mode that is not captured by the scalar horizontal-strain measurement.

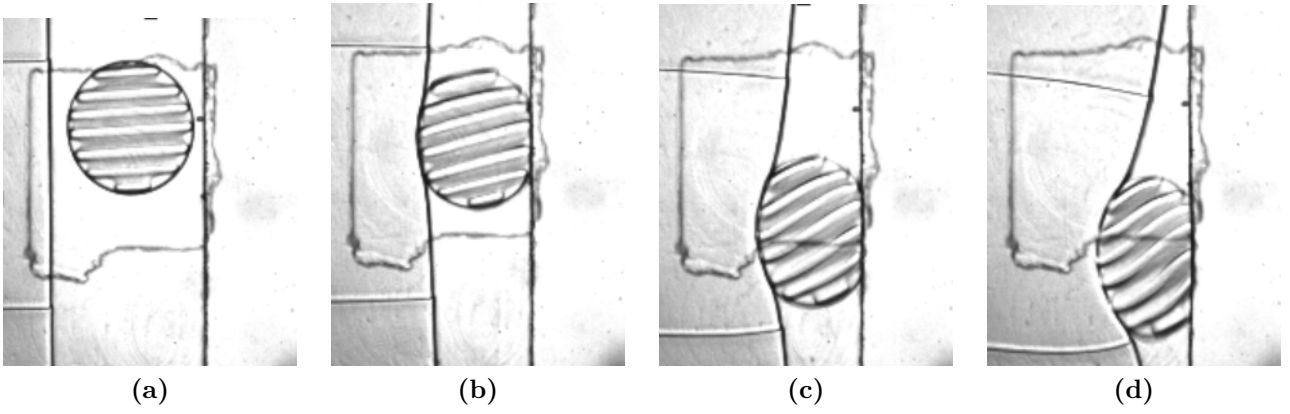


Figure 10: Successive top-view images of the striped cylinder during lateral compression. The initially approximately straight reinforcing stripes progressively rotate and bend, eventually developing S-shaped profiles. Dark regions appearing beside the curved stripes may indicate through-thickness tilting or buckling, although a purely optical origin cannot be excluded from the present top-view images.

5.6 Interpretation and limitations

The final chip supports a conditional relative comparison more strongly than an absolute measurement. The strain itself is sensitive to both the loaded width and the reference diameter. To first order,

$$\Delta\varepsilon_x \simeq -\frac{\Delta D_x}{D_0} + \frac{D_x}{D_0^2} \Delta D_0. \quad (8)$$

For a diameter near 250 pixels, a systematic two-pixel width error changes the measured strain by about 0.008. This is already comparable with a substantial fraction of the separation between the 0.7, 0.8, and 0.9 mean strains. The principal error sources are:

- **Cylinder-boundary tracking.** The external edge is not perfectly resolved in every frame. The contour can be attracted by the pusher interface, the rigid wall, trap texture, or the internal stripes. Wall bounding removes physically impossible penetration on one side, but an error on the pusher-facing boundary still biases D_x . Overlay inspection is therefore indispensable.
- **Reference diameter.** A bias in D_0 affects every frame of one video in the same direction and cannot be reduced by averaging frames. The two videos beginning near contact required an area-based estimate; this assumes approximately conserved top-view area despite possible three-dimensional deformation.
- **Pusher tracking and unknown force.** The pusher indentation coordinate is not a force measurement. Errors in either the detected surface or its far-field reference shift a curve horizontally and cause strains to be compared at different true loads. Calibration against objects of known stiffness or a force sensor would be required for quantitative stiffness values.
- **Imperfect microchip reproducibility.** The same pressure command did not create perfectly identical contact conditions. Small changes in cylinder position, pusher approach, pneumatic lag, ceiling shape, and local contact geometry redistribute deformation between pusher and cylinder. Reusing one chip suppresses chip-to-chip variation but cannot remove run-to-run variation.
- **Pusher history.** Repeated illumination, swelling, relaxation, fatigue, or small permanent displacements can modify the shared pusher during the sequence. A reference cylinder before and after the series, fixed recovery times, and randomised dose order would reveal or reduce this bias.
- **Geometry, confinement, and friction.** The cylinder diameters differed by up to approximately 7% in pixels. Diameter normalisation removes the leading scale effect but not changes in curvature and contact geometry. Nearly incompressible gels also bulge transversely, while friction against the ceiling, floor, wall, and local confinement feature creates a three-dimensional, history-dependent response.
- **Rate dependence.** PEG-based gels and the surrounding fluid can show viscoelastic or poroelastic relaxation. Videos acquired at different frame rates do not by themselves imply different loading rates, but variations in the pressure-ramp duration or waiting time can change the measured strain. Loading, holding, and unloading data should not be mixed.

6 Conclusion

The objective of this internship was first to develop a reliable way of pushing hydrogel cylinders in a deformable microfluidic chip and then to use that capability to compare their relative stiffness. The study began with an existing four-chamber, two-ramp device. Two opposing hydrogel blocks moved as intended, but a free cylinder escaped because the design provided actuation without target confinement. Bonded cylinders, a large ring, and interlocking block shapes confirmed aspects of the motion but did not yield an independent, repeatably loaded sample.

The successful design replaced the symmetric arrangement with one ramp-driven pusher, a fixed wall, and local vertical confinement. The ramp converts downward ceiling motion into lateral pusher translation towards the higher part of the channel, while the wall defines the loading direction. An additional air chamber behind the wall produced no useful retention and was omitted from operation. The aluminium master improved the realisation of ramps and slanted surfaces, and the PDMS-coated glass support allowed hydrogel structures to remain mobile and replaceable.

The quantitative analysis used the direct cylinder boundary rather than an ellipse fit and compared horizontal strain at matched values of a common pusher-deformation coordinate. The plain 0.6 mJ mm^{-2} cylinder was the most deformable plain specimen. The 0.7 mJ mm^{-2} , 0.8 mJ mm^{-2} , and 0.9 mJ mm^{-2} curves were closer and crossed, so their apparent mean order cannot be interpreted as a robust material ranking. The striped 0.6 mJ mm^{-2} cylinder showed the smallest measured mean strain. This is compatible with structural reinforcement, but the cylinder was not geometrically comparable to the plain series and its tracking quality was weaker.

The central outcome is therefore methodological rather than a precise material law. The device repeatedly generated useful compression and produced a visible separation for the most compliant plain specimen within this series. It cannot yet resolve cylinders whose stiffnesses are close. The resulting index is conditional on the tracking, shared-pusher response, confinement, and loading history; it is neither a modulus ratio nor an absolute Young's modulus.

7 Outlook toward quantitative three-dimensional mechanics

The present method provides relative comparisons from the projected deformation of the cylinders, but a quantitative constitutive measurement requires an independent mechanical reference. A first extension would therefore be to characterize the relation between applied pressure, pusher displacement and transmitted force. This could be achieved by compressing reference objects of known stiffness or by integrating a calibrated compliant element into the device. Measurements on several independently fabricated cylinders and chips would then make it possible to separate specimen variability from variations in chip geometry and loading transmission.

A second extension concerns the reconstruction of the internal three-dimensional deformation field. Top-view contour measurements provide only the projected cylinder boundary and cannot distinguish unambiguously between in-plane shear, through-thickness tilting, torsion and local buckling. Volumetric fluorescence imaging of embedded material markers would allow the displacement field $\mathbf{u}(x, y, z)$ to be followed throughout the cylinder thickness. This would also test the validity of the plane-strain approximation imposed by the nominal vertical confinement.

The measured surface and internal displacement fields could subsequently be compared with a finite-strain numerical model including the deformable pusher, cylinder, rigid wall, vertical confinement and contact friction. Combined with force calibration, such a model could constrain the constitutive parameters of the cylinder and determine how much of the observed response originates from material stiffness, contact conditions and three-dimensional geometry. The same approach could ultimately be adapted to cellular spheroids, for which internal rearrangements, spatial heterogeneity and active biological responses make three-dimensional measurements particularly important.

A Annex: operational protocol

A.1 Pre-experiment checklist

1. Verify both fluid ports and every pneumatic line independently. Inspect for bubbles, leaks and imperfections.
2. Prepare the hydrogel from the PEG, PEGDA, and LAP stock solutions in the 60/25/15 volume proportions. Protect the LAP-containing precursor from ambient light.
3. Load the pusher and cylinder masks. Print the pusher first and inspect it for over-polymerisation before printing a cylinder.
4. Prepare the pressure-controller and the video recording software.

A.2 One complete sample cycle

1. Start the recording and start the decompression.
2. If the cylinder leaves the loading region or the pusher develops a large transverse motion, release the pressure, reposition the objects gently, and restart the acquisition.
3. Continue only if the cylinder is vertically confined, touches the fixed wall, and both side boundaries remain visible.
4. Record the complete loading branch and, when a hold is programmed, keep the hold time identical across specimens. Keep the cylinder and pusher interface in the field of view.
5. Unload with the same programmed rate and record recovery for a fixed duration.
6. Flush the cylinder only after full pressure release. Confirm visually that the pusher remains intact, allow a fixed recovery time, and record the sample order before printing the next cylinder.
7. Repeat a reference cylinder at the end of the sequence to quantify pusher and chip drift.

B Annex: image-analysis implementation

B.1 Input preparation

Each recording is exported as a TIFF stack after spatial and temporal cropping in Fiji. The crop must retain the complete cylinder, the fixed wall, the pusher interface, and enough pusher surface above and below the cylinder to infer its unindented reference shape. Several frames before two-sided contact should be retained whenever possible because they provide the most reliable estimate of D_0 . Median-background subtraction and edge enhancement may be used, but the same operations and parameters should be applied to the complete comparison series. All videos to be compared are placed in one input folder and processed by the standalone Jupyter notebook.

B.2 Tracking and measurement sequence

For each stack, the implementation performs the following operations:

1. normalise the orientation so that the fixed wall and pusher occupy consistent sides of the internal coordinate system;
2. detect the initial cylinder and estimate the fixed-wall and far-field pusher positions;
3. use Hough circle and constrained-ellipse searches only to maintain a cylinder localisation window from frame to frame;
4. trace a closed path through the actual cylinder-edge pixels and calculate its direct horizontal span, vertical span, polygon area, and edge-support score;
5. impose the fixed wall as a hard bound on the wall-facing cylinder endpoint after contact, while retaining the measured pusher-facing edge;
6. follow the pusher as a globally continuous edge path, fit its far-field reference above and below the cylinder, and integrate the positive local indentation;
7. identify two-sided contact, retain the monotonic loading branch, and apply the selected strain, area-conservation, and tracking-confidence filters; and
8. interpolate every $\varepsilon_x(\eta_b)$ curve over the shared pusher-deformation interval and compute [equation \(7\)](#).

The notebook displays the initial geometry, representative overlay frames, all strain histories on common axes, the matched-pusher-deformation curves, and the final strain-based table. It also exports frame-level CSV data, an Excel workbook, plots, warnings, and complete diagnostic overlay videos. The overlay is the primary acceptance test: a numerical output is discarded if the displayed cylinder boundary or pusher trace does not follow the visible edge, even when the automatic confidence score is high.

B.3 Code

The analysis notebook is available at github.com/mnifmayad/microfluidic-compression.

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